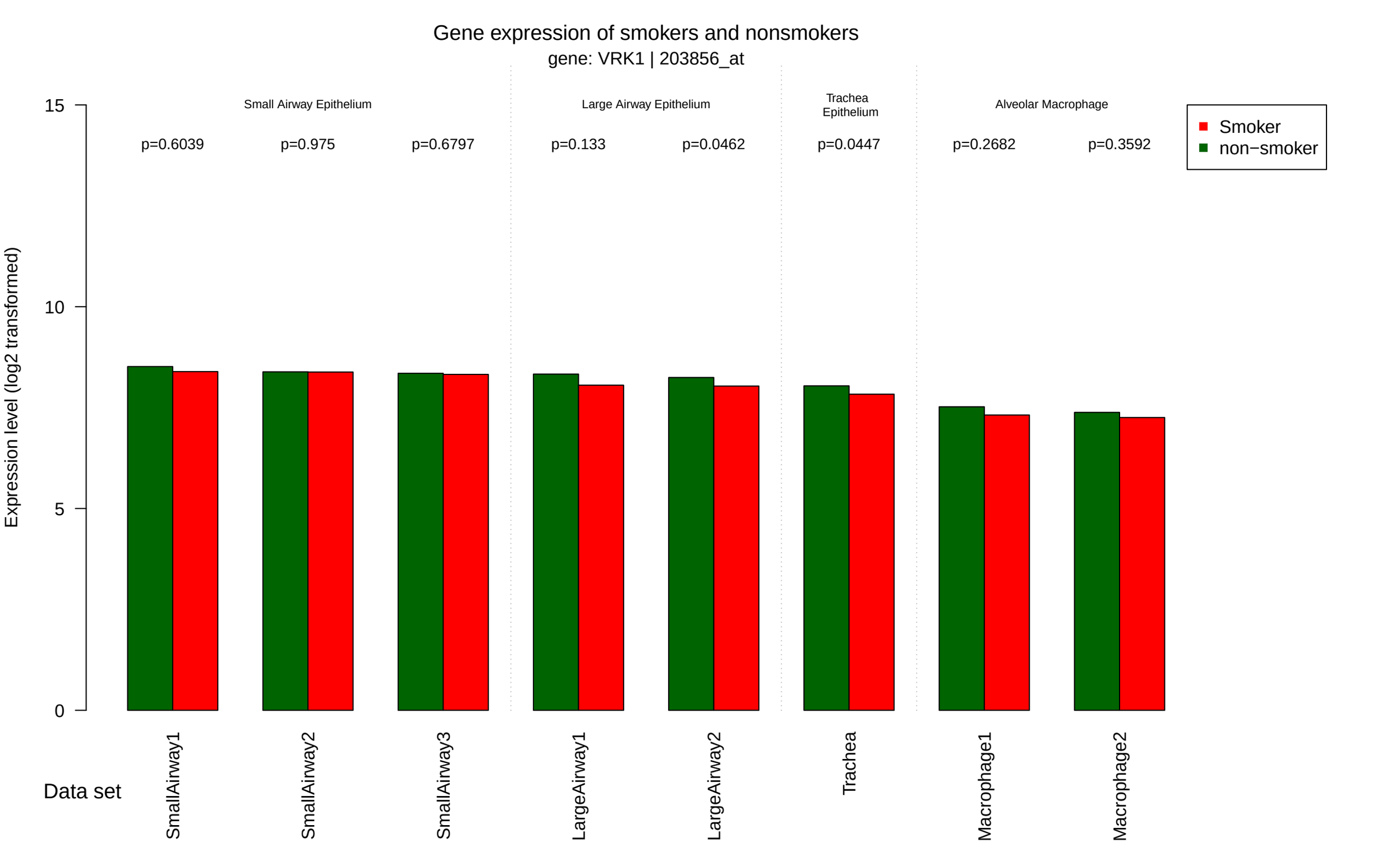
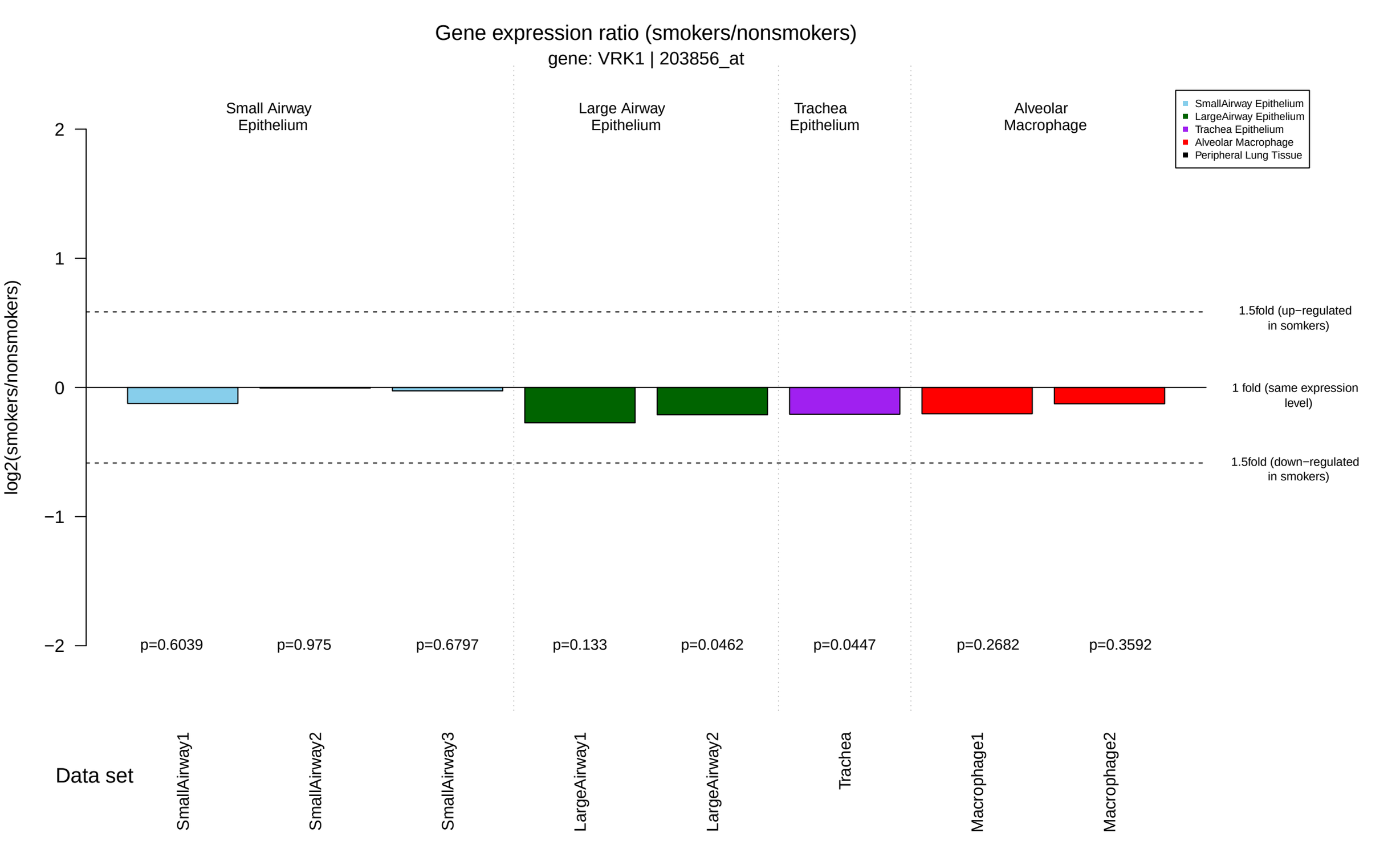
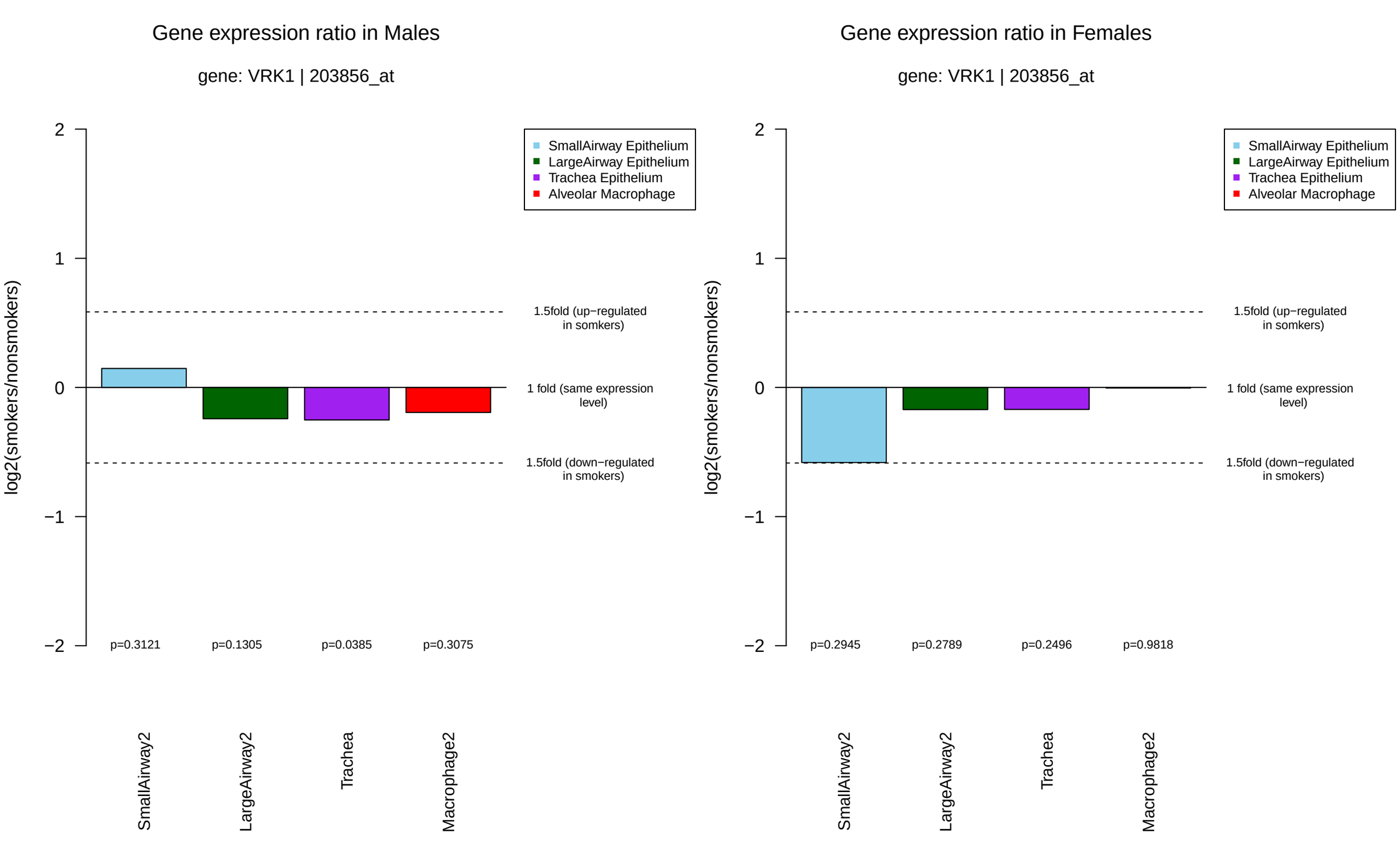
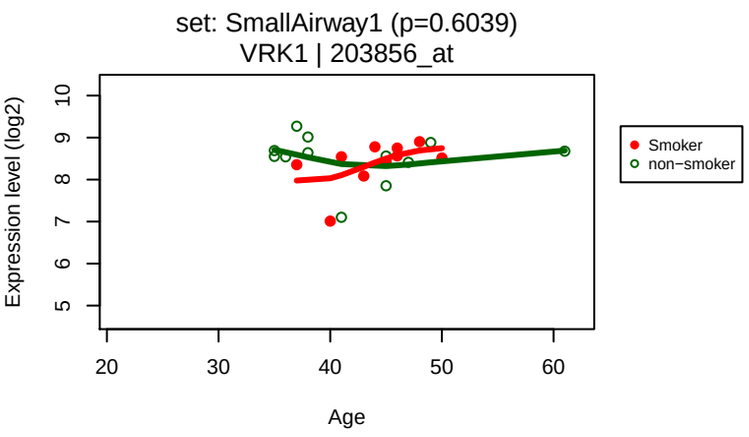
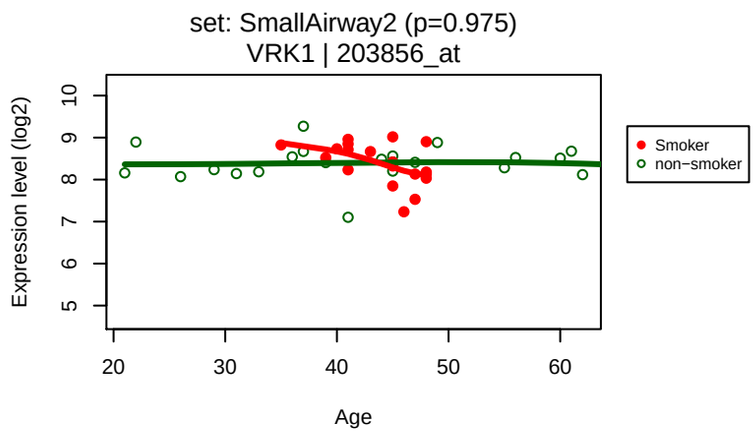
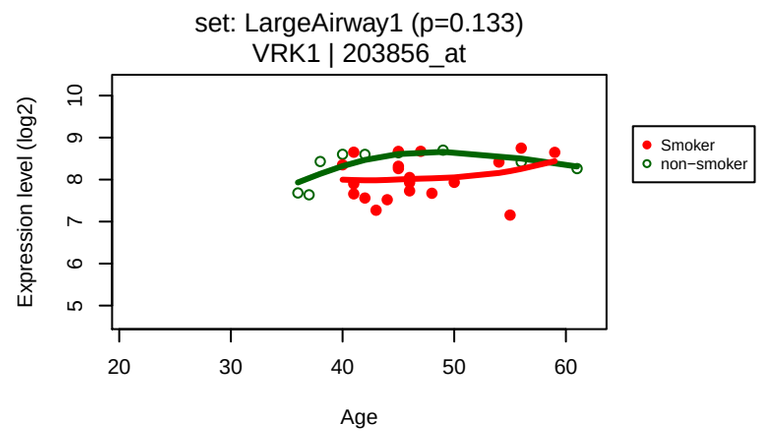

Gene expression of smokers and nonsmokers
gene: VRK1 | 203856_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6039
p=0.975
p=0.6797
p=0.133
p=0.0462
p=0.0447
p=0.2682
p=0.3592
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

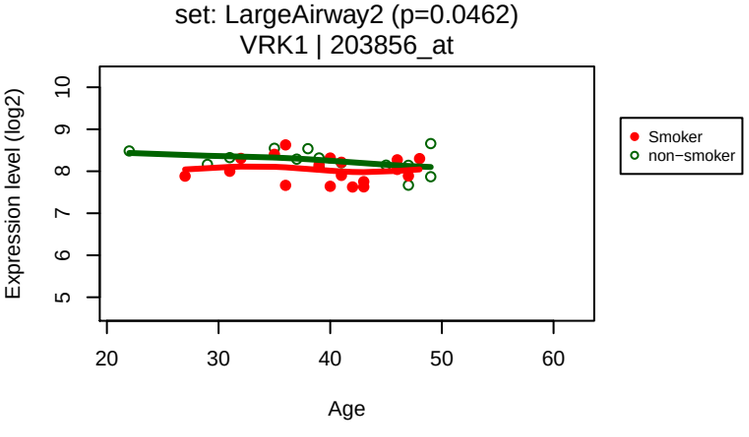

Gene expression ratio (smokers/nonsmokers)
gene: VRK1 | 203856_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6039
p=0.975
p=0.6797
p=0.133
p=0.0462
p=0.0447
p=0.2682
p=0.3592
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

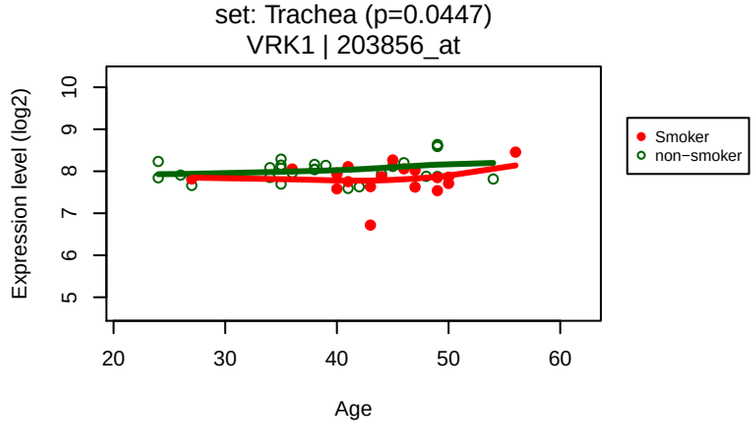

Gene expression ratio in Males
gene: VRK1 | 203856_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3121
p=0.1305
p=0.0385
p=0.3075
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: VRK1 | 203856_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2945
p=0.2789
p=0.2496
p=0.9818
SmallAirway2
LargeAirway2
Trachea
Macrophage2

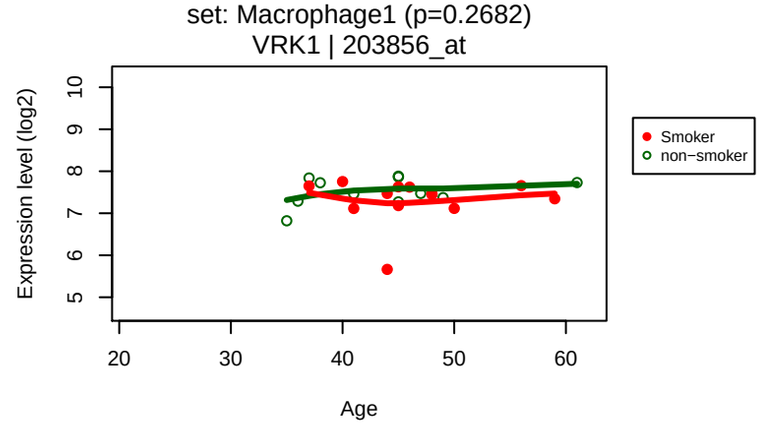

set: SmallAirway1 (p=0.6039)
VRK1 | 203856_at
Expression level (log2)
Age
Smoker
non−smoker

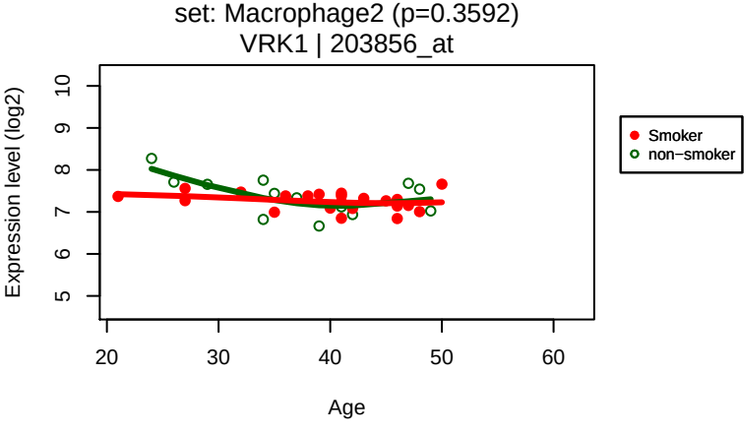

set: SmallAirway2 (p=0.975)
VRK1 | 203856_at
Expression level (log2)
Age
Smoker
non−smoker

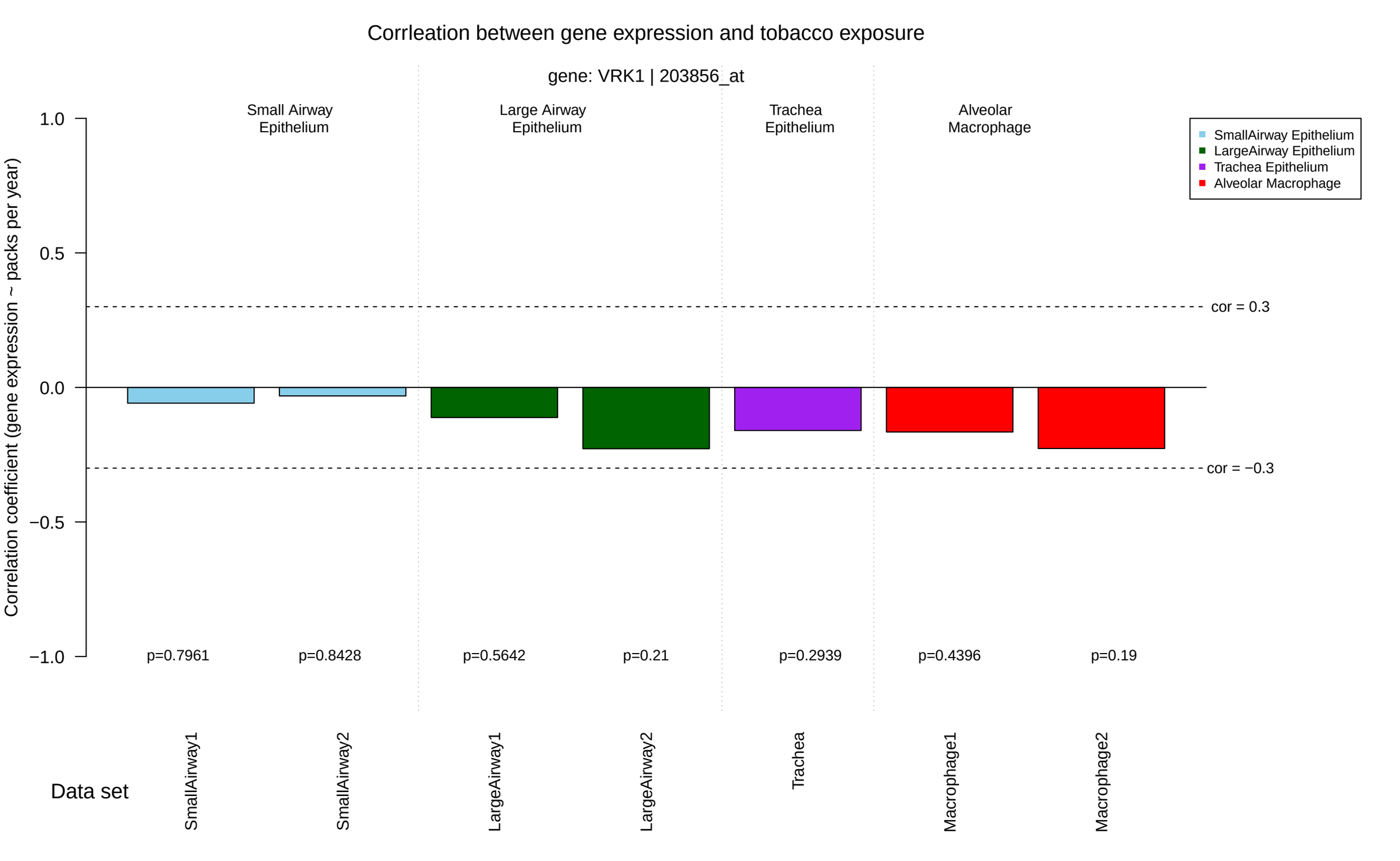

set: LargeAirway1 (p=0.133)
VRK1 | 203856_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0462)
VRK1 | 203856_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0447)
VRK1 | 203856_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2682)
VRK1 | 203856_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3592)
VRK1 | 203856_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: VRK1 | 203856_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7961
p=0.8428
p=0.5642
p=0.21
p=0.2939
p=0.4396
p=0.19
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2